

Research Methodology in CS (CSEG3060_3)

Lecture 6: Research Ethics: foundational principles - honesty, objectivity, integrity, and respect for intellectual property

Unit 3: Ethical Conduct and Literature Studies • Mapped Course Outcome: CO3

Lecture 6: Research Ethics — Foundational Principles: Honesty, Objectivity, Integrity, and Respect for Intellectual Property

Course: Research Methodology in CS (CSEG3060_3) **Program:** B.Tech (Computer Science) **Semester:** 5 **Unit:** 3 — Ethical Conduct and Literature Studies **Lecture Outcome Alignment:** CO3 — Demonstrate understanding of ethical standards governing scholarly research in computer science.

1. Introduction

Research ethics provides the moral and professional compass that guides scholars in the creation, evaluation, and dissemination of knowledge. In computer science — a discipline whose outputs increasingly shape commerce, governance, communication, healthcare, and warfare — the ethical conduct of research carries consequences that extend well beyond the laboratory or the publication page. A single fabricated algorithm, a plagiarized model, or a falsified benchmark can corrupt downstream engineering, mislead policymakers, and erode public trust in the scientific enterprise.

This lecture establishes the four foundational principles that anchor the ethical practice of research:

1. **Honesty**
2. **Objectivity**
3. **Integrity**
4. **Respect for Intellectual Property**

These principles are not abstract ideals. They are operationalized daily in the work of computer scientists — from the design of an experiment, to the writing of a paper, to the reviewing of a peer's manuscript, to the assignment of credit on a multi-author codebase.

The lecture draws on international frameworks, including the **Belmont Report** (1979), the **Singapore Statement on Research Integrity** (2010), the **European Code of Conduct for Research Integrity** (2023 revised edition), and the **ACM Code of Ethics** (2018).

2. Learning Objectives

By the end of this lecture, students should be able to:

- Articulate the meaning and operational scope of each of the four foundational principles.
- Identify concrete ethical dilemmas that computer scientists face in applying these principles.
- Distinguish between honest error and research misconduct.
- Recognize forms of intellectual property — ideas, text, code, data, models — that require respect and attribution.
- Apply ethical reasoning to ambiguous, real-world scenarios.

3. Foundational Principle I — Honesty

3.1 Definition and Scope

Honesty in research refers to the truthful representation of methods, procedures, data, results, and contributions. It is the bedrock upon which the credibility of scientific knowledge rests.

Honesty obliges the researcher to:

- Report results as observed, including negative or unexpected findings.
- Disclose methodologies faithfully, including their limitations.
- Avoid fabrication (inventing data, results, or experiments).

- Avoid falsification (manipulating research materials, equipment, or processes, or altering or omitting data).
- Report errors promptly and correct the record when mistakes are discovered.

3.2 Why Honesty Matters in Computer Science

Computer science research is largely driven by *artifacts* — software, hardware, datasets, benchmarks, and trained models. The reproducibility of these artifacts depends on honest reporting. Consider:

- **Benchmark performance:** A 2023 audit of machine learning papers (Oakden-Rayner et al.) found that selective reporting of metrics — reporting only the best of many runs — artificially inflates claimed performance gains. This constitutes a form of *selective honesty*: technically true statements that omit context, misleading readers.
- **Algorithm novelty:** Claiming an algorithm is "novel" when it is essentially a renamed prior method is dishonest even though it is non-trivial to detect.
- **Data integrity:** Reporting on a cleaned subset of a dataset without disclosing exclusions can lead reviewers and readers to draw invalid conclusions.

3.3 Distinguishing Honest Error from Misconduct

Not every wrong result is misconduct. The distinction rests on **intent** and **transparency**:

Dimension	Honest Error	Misconduct
Intent	Unintentional, accidental	Willful or reckless
Response to discovery	Open acknowledgment, correction	Concealment, denial
Documentation	Reproducible records of process	Inconsistent, missing
Frequency	Sparse and explainable	Pattern of irregularities

A useful heuristic is the **"Fabrikation–Falsifikation–Plagiat" (FFP) framework** widely adopted in research integrity policy. Honesty is the positive counterpart that prevents FFP in the first place.

3.4 Operational Practices

- Maintain a tamper-evident **research log** (version-controlled lab notebook).
- Preserve **raw data** with metadata indefinitely where feasible.
- Adopt **pre-registration** for confirmatory studies.
- Use **double-blind review** norms to deflate incentive for selective reporting.

4. Foundational Principle II — Objectivity

4.1 Definition and Scope

Objectivity is the commitment to let evidence — not personal interest, ideology, prior commitment, or external pressure — drive the design, analysis, and interpretation of research. It is the discipline of allowing the data to speak, even when the message is unwelcome.

Objectivity obliges the researcher to:

- Design experiments and evaluations in ways that can in principle falsify a hypothesis.
- Avoid conflicts of interest, or disclose them when unavoidable.
- Distinguish empirical observation from interpretation.
- Submit one's claims to peer scrutiny rather than defend them rhetorically.

4.2 The Challenge of Objectivity in Computer Science

Computer science is often perceived as "objective" because it deals with formal systems. Yet every layer of an empirical CS study — from dataset choice, to hyperparameter selection, to evaluation metric — is shaped by human judgment. Objectivity is a *regulative ideal*, requiring continuous effort.

Consider three pressure points:

(a) Confirmation bias in algorithm design. A researcher who has emotionally invested in a proposed algorithm may unconsciously design benchmarks that favor it. Robust objectivity requires running baselines under matched conditions, using standard suites, and reporting performance across the full

distribution of inputs, not just cherry-picked cases.

(b) Conflicts of interest. Industry funding, equity stakes, consulting relationships, and personal relationships can compromise the disinterested stance that objectivity requires. Disclosure is necessary but never fully neutralizing; readers should interpret funded work accordingly.

(c) P-hacking and significance hunting. Although computing research rarely uses null-hypothesis significance testing in the biomedical sense, analogous risks exist. Tweaking architectures, hyperparameters, random seeds, or preprocessing pipelines until a result "works" corrodes objectivity even when the technique is conventional.

4.3 Statistical and Mathematical Literacy

Objectivity is, in part, a competence. Consider a claim that a new compiler optimization yields a mean speedup of 12% over a baseline. Without confidence intervals, standard deviations, hardware variance, and sample-size justification, the claim is technically a number but not an empirical finding. Reporting performance gains rigorously often requires:

$$\text{Cohen's } d = \frac{\mu_{\text{new}} - \mu_{\text{baseline}}}{\sigma_{\text{pooled}}}$$

Cohen's $d = \sigma_{\text{pooled}} \mu_{\text{new}} - \mu_{\text{baseline}}$

$$95\% \text{ CI: } \bar{x} \pm t_{0.025, n-1} \cdot \frac{s}{\sqrt{n}}$$

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$\sqrt{\frac{s^2}{n}}$

The first expression gives effect size; the second gives uncertainty. Together, they allow readers to assess the claim on its merits rather than on the author's enthusiasm.

4.4 Operational Practices

- Use **pre-registered** experimental protocols where appropriate.
- Apply **blinding** in evaluation: the person running benchmarks should not know which system is the proposed one.
- Disclose all **conflicts of interest** explicitly.
- Encourage **adversarial collaboration** — designing experiments that a skeptic would accept.

5. Foundational Principle III — Integrity

5.1 Definition and Scope

Integrity is the consistent and coherent application of ethical principles across the entire research lifecycle. While honesty pertains to specific statements and data, integrity pertains to the *character* and *consistency* of the researcher. A researcher of integrity acts ethically even when no one is watching, when peer pressure pushes the other way, or when expedience tempts compromise.

Integrity subsumes:

- Adherence to institutional and professional standards.
- Consistency between what one says publicly and what one does privately.
- Willingness to take ethical stands that may be costly — for instance, retracting one's own work, or whistleblowing on a colleague's misconduct.
- The disposition to ask "Is this the right thing to do?" prior to every significant research decision.

5.2 Integrity in Multi-Author and Multi-Site Work

Modern CS research is collaborative. Integrity must scale across teams, institutions, and nations. This raises practical questions:

- **Authorship attribution.** The **CRedit (Contributor Roles Taxonomy)** provides 14 contributor roles (conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualization, supervision, project administration, funding acquisition, resources). Misattribution of these roles is an integrity violation.
- **Gift authorship and ghost authorship.** Including someone who did not meaningfully contribute, or omitting a contributor whose work materially shaped the paper, are both breaches of integrity.
- **Order of authorship.** The convention in CS — usually alphabetical — sometimes masks the actual contribution order. Integrity requires transparent disclosure when the alphabet misrepresents the work distribution.

5.3 Integrity in Mentoring and Supervision

Computer science research is heavily mentored. A PhD advisor's integrity directly shapes a student's professional formation. Practices that compromise integrity include:

- Pressuring students to keep the lab's "story" consistent in reviews.
- Accepting authorship in exchange for funding or lab resources.
- Failing to credit students whose code, datasets, or experiments populate papers.

5.4 Whistleblowing and the Costs of Integrity

Whistleblowers often pay a price — retaliation, social ostracism, stalled careers — yet whistleblowing remains essential to the integrity of the scientific system. Institutional **Ombudspersons**, confidential reporting channels, and protective policies (e.g., the U.S. Whistleblower Protection Act, EU Directive 2019/1937) exist to mitigate these costs. B.Tech graduates entering industry research may one day face such choices, and a mature understanding of integrity prepares them to act.

5.5 Operational Practices

- Adopt transparent authorship frameworks (CRediT).
- Establish lab agreements on data ownership and publication.
- Document difficult decisions and their ethical reasoning.

6. Foundational Principle IV — Respect for Intellectual Property

6.1 Definition and Scope

Respect for intellectual property (IP) acknowledges that ideas, expressions, methods, data, and code have authors, and that these authors have legitimate claims over attribution, and in some cases over use and commercialization. This principle is the bridge between ethical conduct and legal frameworks (copyright, patents, trade secrets), but it is also broader: respect for IP includes moral acknowledgments that go beyond what the law strictly requires.

6.2 Categories of Intellectual Property in CS Research

A. Textual IP — plagiarism and attribution

Plagiarism is the appropriation of another person's ideas, processes, results, or words without giving proper credit. Two forms are distinguished in the literature:

- **Verbatim plagiarism** — copying sentences or paragraphs verbatim without quotation marks or citation.
- **Idea plagiarism** — paraphrasing another scholar's argument so closely that the structure and substance remain the source's, with only superficial word changes.

Computer science papers are densely argumentative; even a sentence lifted without quotation constitutes misconduct. The threshold for attribution is much lower than in many other disciplines.

B. Software and code IP

Source code is both functional and expressive. Respect for code IP includes:

- Not copying open-source code without honoring license terms (GPL, MIT, Apache, BSD all impose distinct obligations).
- Not re-implementing a protected algorithm's expressive structure after reading another team's code.
- Citing software libraries used in experiments.
- Honoring reproducibility norms: when publishing, deposit code where possible under a permissive license.

C. Data IP

Datasets are research outputs. Their use carries obligations:

- Citing the dataset creators.
- Respecting licenses (Creative Commons variants, dataset-specific terms).
- Not redistributing restricted-access datasets through side channels.
- Respecting the consent terms under which human-subjects data were collected.

D. Algorithmic and methodological IP

Even unpatented algorithms have authors. Naming and citing the origin of a method is part of research hygiene. The community-standard convention is to use the originator’s name: RSA (Rivest–Shamir–Adleman), Dijkstra’s algorithm, Adam optimizer (Kingma and Ba), Transformer (Vaswani et al.), ResNet (He et al.).

E. Trademark and brand IP

Naming a system after a competitor’s trademarked product, or using screenshots of proprietary software beyond fair-use bounds, can infringe trademark and copyright. Even when legally defensible, the practice may be unprofessional.

6.3 The Legal–Ethical Boundary

IP law varies by jurisdiction and is broader than research ethics in some respects (e.g., patent law) and narrower in others (e.g., many forms of idea plagiarism are legal but unethical). A useful schematic:

Dimension	Legal	Ethical
Verbatim copying of text without citation	Generally not illegal (small amounts, fair use)	Unethical
Re-implementing an algorithm from reading prior code	Generally legal	Often unethical
Using a copyrighted figure in a review paper	Often permitted under fair use	Still requires citation
Falsifying results	Not typically illegal	Unethical; sometimes illegal (e.g., federal grants)
Stockpiling research data for years	Generally legal	May be unethical if it blocks community progress

The ethical principle of respect for IP operates independently of, and often more strictly than, the legal minimum.

6.4 Operational Practices

- Use **citation managers** (Zotero, Mendeley, EndNote) to track every idea, method, and quote.
- Run submitted papers through **plagiarism-detection software** (Turnitin, iThenticate) before submission.
- Maintain a **license awareness checklist** for every dependency used in research code.
- When paraphrasing, ensure the structure as well as the wording is your own.

7. Case Studies

Case 7.1 — The Case of the Trimmed Benchmarks (Composite Scenario)

A graduate student is benchmarking a newly proposed neural architecture against five baselines. Initial runs show the proposed architecture underperforming three of them. After several re-runs and one round of hyperparameter tuning, the proposed architecture outperforms all baselines. The student reports the final results without disclosing the failed early runs.

- **Which principle is implicated?** Honesty, and partially Integrity.
- **Is this misconduct?** Yes. Selective reporting of successful runs while omitting failed ones distorts the literature.
- **Best practice:** Report the full set of runs, document hyperparameter search protocol, and use statistical tests for the comparison.

Case 7.2 — The Borrowed Paragraph

A B.Tech student writes a literature review. They take a paragraph from a published survey, replace a few words with synonyms, and insert it into their own paper with no citation.

- **Which principle?** Respect for IP (plagiarism).
- **Severity:** High — verbatim or near-verbatim appropriation of text.
- **Best practice:** Reread the source, put it aside, and write the idea in your own framing, citing the source properly; or quote verbatim with quotation marks and citation.

Case 7.3 — The Ghost in the Code

A senior PhD student completes the implementation and evaluation of a method. Their advisor adds a paragraph to the paper, places both names on the byline, and submits. The student is in the third year and is told that “first authorship will be sorted out later.”

- **Which principle?** Integrity (authorship credit).
- **Severity:** Potential. CRediT roles and transparent authorship agreements are the remedy.
- **Best practice:** Agree on authorship before submission, using a documented authorship policy consistent with the target venue's standards.

Case 7.4 — The Open-Source Re-Use Trap

A researcher forks a GPL-licensed library, modifies it, and uses it inside a closed-source commercial product. They argue that they "wrote the changes themselves" and so are not bound by the GPL. The library was a derivative work of the original.

- **Which principle?** Respect for IP; also Integrity.
- **Severity:** Legal exposure is high; ethical breach is clear because the license terms were explicit.
- **Best practice:** Maintain a license inventory for every dependency; consult legal counsel where compliance is unclear.

Case 7.5 — The Convenient Control Experiment

An industry-sponsored team compares their new system to the previous state of the art. They omit a recently published open-source baseline that outperforms their system on standard metrics, because "we ran out of time." The omission is silent; reviewers assume the comparison is exhaustive.

- **Which principle?** Objectivity.
- **Severity:** High. Omitting a known relevant comparator biases the contribution narrative.
- **Best practice:** Disclose all known relevant comparators; explain why any were excluded.

8. Synthesis — How the Four Principles Interact

The four principles are not independent. A single act of conduct can implicate all four.

- **Fabricating data** is dishonest, lacks objectivity in the sense that evidence is decoupled from inference, reflects a failure of integrity, and disrespects the IP of the community whose trust is being betrayed.
- **Plagiarizing** disrespects IP, is dishonest about authorship contribution, compromises integrity of the scholarly record, and skews the objectivity of the literature by inflating apparent support for claims.

A useful **diagnostic workflow** when facing a borderline decision:

1. **Honesty test** — Can I state this plainly, including what is less flattering?
2. **Objectivity test** — Has the evidence, not my preference, driven my conclusion?
3. **Integrity test** — Would I be comfortable if this decision were reported in the campus newspaper?
4. **IP-respect test** — Have I given proper credit to sources, code, data, methods?
5. **Public-reason test** — Can I justify this decision in a forum of respected peers?

If all five pass, the decision is almost certainly ethical. If any fails, further reflection — or consultation with an advisor or ethics officer — is warranted.

9. Summary

This lecture has unpacked the four foundational principles of research ethics:

- **Honesty** demands truthful reporting of methods, data, and findings; it is the antidote to fabrication, falsification, and selective reporting.
- **Objectivity** demands that evidence, rather than interest, drive inference; it requires the management of conflicts of interest, the avoidance of confirmation bias, and the discipline of methodological rigor.
- **Integrity** demands consistency of ethical conduct across the entire research lifecycle, including authorship, mentorship, and the willingness to defend the truth when it is costly to do so.
- **Respect for intellectual property** demands attribution of ideas, text, code, data, and methods, together with compliance with the licenses under which they are shared.

These principles operate together; weakness in one weakens the system as a whole. In computer science — a field whose outputs scale rapidly into society — adherence to these principles is not optional. It is constitutive of the discipline's claim to scientific status.

10. Exam Preparation and Practice

10.1 Likely Short-Answer Questions

1. Define **research integrity** and distinguish it from **research honesty**, with one example from computer science.
2. Explain why **selective reporting of benchmark results** is a violation of honesty even when no data are fabricated.
3. Distinguish **idea plagiarism** from **verbatim plagiarism** and provide one example relevant to a literature review.
4. Describe the role of the **CRedit taxonomy** in supporting integrity in multi-author work.
5. Why is the **legal-ethical boundary** important when discussing respect for intellectual property? Provide one example.
6. State the diagnostic workflow (five tests) and apply it to a hypothetical scenario of your choosing.

10.2 Likely Long-Answer / Essay Questions

1. "**Objectivity is a regulative ideal, not a state.**" Discuss with reference to conflicts of interest, confirmation bias, and benchmark design in contemporary machine learning research.
2. Analyze a case where each of the four foundational principles is implicated. Discuss the institutional mechanisms that can catch or correct such a violation.
3. Compare the protection of **textual IP**, **software IP**, and **data IP** in computer science research. What license-attribution practices should a B.Tech project adopt?

10.3 Applied / Case-Analysis Prompts

- *You are a third-year B.Tech student co-authoring a paper with a peer who has removed your name from the byline because they disagree with your recent methodological criticism. How do you proceed, and which principles are at stake?*
- *Your advisor pressures you to report benchmark results from a partial dataset because "the full dataset will take too long." Identify the principles and propose a constructive course of action.*
- *You find an elegant open-source implementation that does almost what you need for a publication. Compare three options: (a) fork and modify citing the source, (b) reimplement from scratch using only the paper's description, (c) use as-is and relicense. Discuss the IP and integrity implications of each.*

10.4 Concept-Map Checklist for Revision

- Honest Error vs Misconduct (intent, response, documentation)
- Fabrication — Falsification — Plagiarism (FFP framework)
- Selective reporting vs. Negative-results reporting
- Objectivity vs. Confirmation bias vs. Conflict of interest
- Integrity vs. Honesty — character vs. specific statement
- Authorship: gift, ghost, alphabetical, CRedit roles
- IP categories: text, code, data, algorithms, trademarks
- License compliance: GPL, MIT, Apache, BSD, CC variants
- Diagnostic workflow: Honesty, Objectivity, Integrity, IP, Public-reason

10.5 Recommended Further Reading

- Singapore Statement on Research Integrity (2010).
- European Code of Conduct for Research Integrity, revised edition (2023, ALLEA).
- ACM Code of Ethics and Professional Conduct (2018).
- Resnik, D. B., *What Is Ethics in Research & Why Is It Important?* (National Institute of Environmental Health Sciences, online resource).
- Steneck, N. H., *Introduction to the Responsible Conduct of Research* (Office of Research Integrity, U.S. Department of Health and Human Services).
- For CS-specific concerns: Nuts and Bolts of Machine Learning reproducibility surveys, e.g., **Pineau et al., "Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program)", *Journal of Machine Learning Research*, 2021.**

End of Lecture 6 Notes.